

# 2

## ER MODEL

### 2.1 Introduction

Entity relationship diagram used to represent diagrammatic design (High level design) of DB.

### 2.2 Main components in ER Diagram

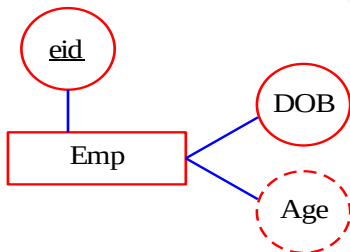
- (i) Attributes
- (ii) Entity sets
- (iii) Relationship sets

#### (a) Entity sets

It is a set of entities of the same type denoted by a rectangular box in ER diagram. Entity can be identified by a list of attributes which are placed in ovals.

Represent by : 

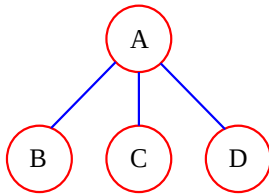
**Example :**



#### (b) Attributes

- (i) Attribute : 
- (ii) Key attribute : 
- (iii) Derived attribute : 

(iv) Composite attribute: Attribute which can be represented as two or more attributes.

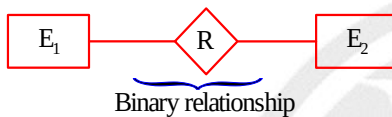


(v) Multivalued attribute:

(c) **Relationship set:** It is used to relate two or more entity set.

Represented by :

**Example:**



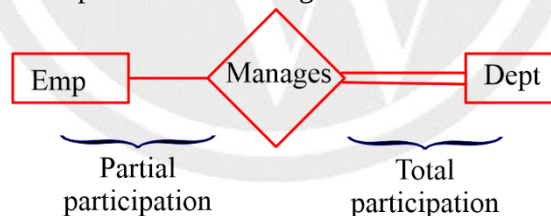
## 2.3 Participation

- If every entities of entity set are participated with relationship set then it is total participation (100% participation) otherwise it will be partial participation (< 100% participation)

**Example :**

Consider Emp and Dept entity set.

Manages relationship set such that each dept must have manager.

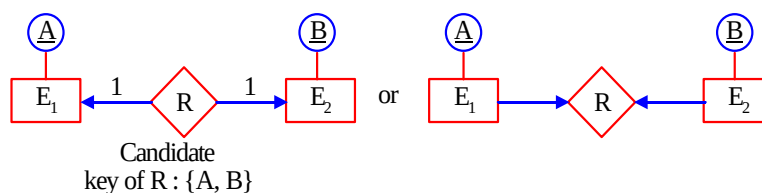


## 2.4 Mapping [Cardinality constraints of relationship set]

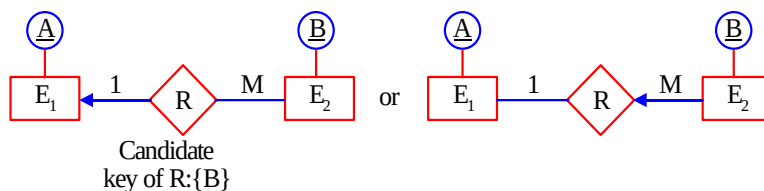
One mapping : Atmost one (0 or 1)

Many mapping : 0 or more (0 ..... \*)

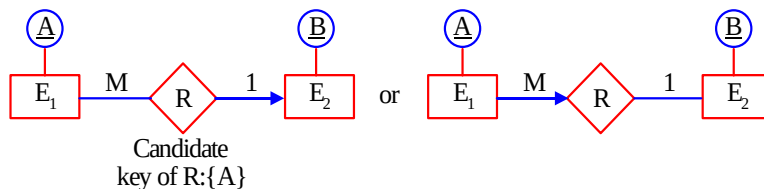
### 1. Binary Relationship Mapping (One : One)



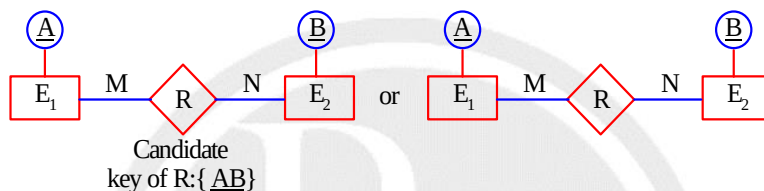
### 2. Binary Relationship Mapping (One : Many)



### 3. Binary Relationship Mapping (Many to One)



### 4. Binary Relationship Mapping (Many to Many)



## 2.5 RDBMS Design of Given ER Diagram

(For binary relationship)

### 1. Partial participation on both side of binary relationship

- **One to Many** : Merge relationship set towards many side. So, 2 relational tables.
- **Many to one** : Merge relationship set towards many side. So, 2 relational tables.
- **One to one** : Merge relationship set any one side. So, 2 relational tables.
- **Many to Many** : Separate table for each entity set and relationship set. so, 3 relational tables.

### 2. Full participation on “one” side of many to one relationship

Merge the entities and relationship set into single relational table. So, 1 table.

### 3. Full participation on “Many” side of Many-to-one relationship

Merge relationship set towards many side. So, 2 relational tables.

### 4. Full participation on any “one” side in one-to-one relationship

Merge the entity sets and relationship set into single table. So, 1 table.

### 5. Full participation on any “Many” side of Many-to-Many relationship

Merge relationship set towards any “Many” side of relationship. So, 2 table.

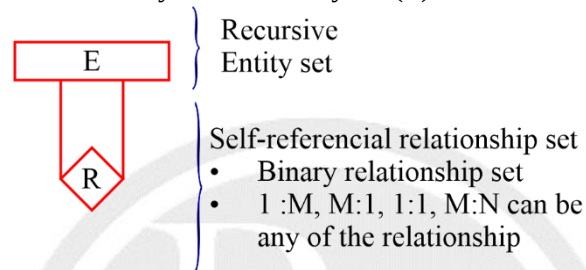
## 6. Full participation on both side of relationship

- |                                                                                                      |   |                                                                                                       |
|------------------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• 1:1</li> <li>• 1:M</li> <li>• M:1</li> <li>• M:N</li> </ul> | } | <p>Merge the entity sets and relationship set into single relational table so, 1 relational table</p> |
|------------------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------------|

## 2.6 Self-Referential Relationship Set

(Recursive entity set)

Entities of entity set (E) related to some other entity of same entity set (E).



## 2.7 Weak Entity Set and Weak Relationship Set

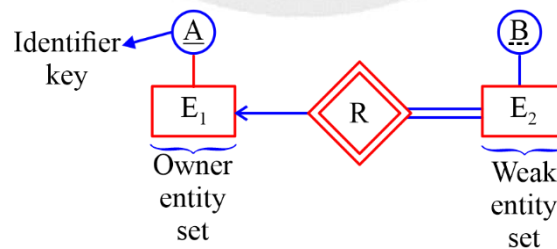
The entity set with no key. (Attributes of weak entity sets are not sufficient to differentiate entities uniquely).



### Points:

- For each weak entity set there must be owner entity set, which is strong entity set.
- Relationship set between weak entity set and identifier entity set is also “weak relationship set”.
- The participation towards weak entity set end must be “total participation”.
- The mapping between identifier entity set and weak entity set must be one : many (1 : M)

### Example:



### Note:

Weak entity set and multivalued attributes allowed to represent in ER diagram but not allowed in RDBMS table.

